JNTUH USED 23-05-18AM Set 2

Code No: 117EF **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, April/May - 2018
MACHINE LEARNING
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART- A

(25 Marks)

1.a) What is meant by machine learning? [2]
b) What are the key ideas of statistics which influence machine learning? [3]
c) Illustrate a perceptron. [2]
d) What is the use of Occam's razor? [3]
e) State central limit theorem. [2]
f) What is the procedure for calculating confidence intervals for discrete-valued hypotheses? [3]
g) What are the advantages of instance based methods? [2]
h) Make a comparison of eager learners and lazy learners. [3]
i) Define weakest preimage. [2]
j) What is crossover operator of genetic algorithm? [3]

PART-B

(50 Marks)

2. Illustrate basic design issues and approaches to machine learning by considering a program to learn to play checkers. [10]

OR

3.a) Discuss the limitations of FIND-S algorithm.
b) Write candidate-elimination algorithm using version spaces. [5+5]

4.a) Discuss the issue of over fitting the training data with example.
b) Contrast the hypothesis space search performed by ID3 with candidate elimination algorithm. [5+5]

OR

5. Explain back propagation algorithm for training a neural network for learning task.[10]

6.a) Describe a procedure to estimate the difference in error between two learning methods.
b) How does Bayesian method outperform decision tree? What is the practical difficulty in applying Bayesian methods? [5+5]

OR

7. What is Bayesian belief network? How is it trained using expectation maximization algorithm? Explain with an illustrative example. [10]

8. Demonstrate distance-weighted k-nearest neighbor algorithm with an example dataset. [10]

OR

9.a) "Radial basis function networks provide a global approximation to the target function, represented by a linear combination of many local kernel functions." Substantiate the statement.

b) What is case based reasoning? Why is it known as lazy learner? [5+5]

10. What is the need of fitness function in genetic algorithm? Explain with suitable example. [10]

OR

11.a) Make a comparison of analytical learning and inductive learning.

b) Discuss explanation based learning algorithm PROLOG-EBG. [5+5]

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